Project Milestone P2

Analysis and High-Level Design

BookPack

Team	Team 7, CE 3354.003
Document version	1.8 (baseline for P2; grows from your P1 version 0.8)
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0 Revision History, Contribution Record and Changes Since P1

All three tables below are graded. The contribution record carries the individual 20% of this milestone and must agree with your repository history.

Version	Date	Author	Summary of change
0.8	10/18/206	Ahmet	Added sections 2.1, 2.5 and 5
1.0	11/7/2026	Patrick	Added documentation README
1.1	11/8/2026	Patrick	Added section 10
1.2	11/8/2026	Ahmet	Added Section 6, ADR-01/03, UC-03, Section 11
1.3	11/8/2026	Peter	Added Stakeholder Interview
1.4	11/8/2026	Patrick	Updated section 1, section 2, and section 10
1.5	11/8/2026	Afeef	Added UC-01 expansion (4.1,4.2,4.3), including the valuable book exception
1.6	11/8/2026	Jasmine	Added 5.2.1
1.7	11/8/2026	Jasmine	Added Section 9
1.8	11/8/2026	Lucas	Completed section 5.1 with Peter and completed section 5.3

0.1 Individual contribution record (P2)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Nicholas Suther	Section 3 carried forward with CH-01 to CH-03 and FR-16 (CH-05); UC-02 expansion with its 4.1 and 4.3 rows and open question Q8; Section 8 spike S-01 (simulated scale, reader, test runner, measured results, README); ADR-02 scale integration and its 7.2 trade-off row; 0.2 change rows for FR-08, FR-09 and FR-03/Table 3.1	/spikes/S-01-scale-read/ , /docs/p2/; commits ce38eaf, 650e120, 392fe56, 79296ba; PR #10 (Section 3; commits d8280b7, 5fb9f4e, 4802d2b, 71906d9, e83be7e, 9636d8d); PR #15 (UC-02; commit 10841bd)	PR #6 and PR #7 (changes requested, then approved; both merged), PR #8 (approved)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Peter Yu	Stakeholder Interview	PR#13 (074e810)	PR#9 (38abf27), PR#16(648d119), PR#10 comment, PR#19(ab06c44), PR#14(c8406fb), PR#18(d2b8f53), PR#20(1744b12)
Lucas Pham	Completed section 5.1 and completed 5.3	PR#19 (Section 5.1) PR#20(Section 5.3) commits 8129948, 5344553, f675aef, 1a9cafc, 74de6b2, 137280e, 5b1807	PR#18 (approved)
Patrick Paul	Wrote and revised P2 Section 1 (purpose, scope, references R1–R5, FR-08/09 and CH-03 consistency), Section 2 (stakeholders, P1 constraints C1–C5 with type/source and CON-01–03 traceability, corrected FR-13), and Section 10 (working agreement, P3 milestones, risks R1–R7, repository/submission procedures, and AI-use disclosures). Added docs/p1/README.md in response to P1 instructor feedback. Synchronized Sections 1–2 in the shared P2 Google Doc. Final whole-document and PDF review remain pending.	PR #6 (Section 1; commit 9af75c3; merged); PR #7 (Section 2; commit 21c8dd5; merged); PR #8 (Section 10; merged as d937ba5); PR #9 (docs/p1/README.md; merged); PR #11 (Section 10 editorial follow-up; merged). GitHub account: sirbottoms98.	PR #10 — two COMMENTED reviews (Nov. 8, 2026): requested complete QR/CON coverage and consistent CH IDs; identified physical packaging-fit questions, FR-03/FR-10 dimension persistence gaps, and FR-16 valuable-book flag/acceptance-test gaps.

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Ahmet Keles	Section 6 (6.1–6.6) with component and deployment diagrams; ADR-01 and ADR-03 and their 7.2 rows; UC-03 expansion (4.2), its 4.3 review rows, 5.2.2 scenario table and sequence diagram; section 11 open questions	/docs/p2/ (section-6-architecture, adr-01-layered-style, adr-03-configurable-rules, uc-03-inspect-flagged-book and section-11-open-questions .md files), /docs/p2/diagrams/; commits f58a959, a112ad6, e61d011, 81eeb71, cc302b0, 70c3d7e, f5ad40d, e033cd4; PR #12	PR #13 (approved)
Afeef Zarraf	Use case analysis: UC-01 process book for shipment, which I wrote in p1(3,4), expanded on p2 in 4.2 with step (0), turns (1) to (10) and seven alternate flows, such as the valuable book exception (FR-16, CH-05); its 4.1 row, and the UC-01 rows of 4.3 self review log.	PR#18(Section 4 UC-01; docs/p2/uc-01-process-book-for-shipment.md; commit 557d78e). Github account: afeef06	Merge pull request #17 from PeterYuGit/p2/jasmine-section 5.2.1 (approved)
Jasmine Tran	Section 9 (9.1, 9.2) with quality requirements verification mapping and measurement validation, P2 traceability reviews. Section 5.2.1 UC-02 (non-trivial steps, scenario table, and sequence diagram)	PR#14 /docs/p2/section-9-traceability merged as c8406fb, PR#17 /docs/p2/diagrams, /docs/p2/uc-02-override-packaging.md merged as 9e84659	PR#15 (approved)

0.2 Changes since the P1 baseline

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	FR-08 acceptance criterion changed to: “Entered manually, 5.0 lb is not flagged and 5.1 lb is flagged. Read from the scale, any book over 5.05 lb is always flagged; a book within the scale’s ±0.05 lb tolerance of 5.0 lb may be flagged.	Spike S-01: scale noise caused a 5.00 lb book to read just over 5.0 lb in 5 of 15 trials, so the old criterion cannot pass reliably. Extra inspection is the safe direction to be wrong in.	ADR-02 cites this change; Q9 (flag limit) is still open with the bookstore.	11/08/26
CH-02	FR-09 priority raised from SHOULD to MUST.	FR-09 enforces the same 70 lb limit as CON-01, which is non-negotiable (USPS); the two priorities contradicted each other.	None; FR-09 is already realized in UC-01 step 4 and alternate flow 4a.	11/08/26
CH-03	Table 3.1 note added: classification uses weight and thickness only. FR-03 kept, because length and width are still recorded for the shipping record and shown to the inspector (FR-14). UC-01 step 6 reworded from “dimension thresholds” to “the weight and thickness thresholds in Table 3.1”.	Review found that FR-03 collects length and width no rule used, and UC-01 step 6 described classification differently from Table 3.1.	UC-01 in section 4.2 and the domain model must use the same wording.	11/08/26
CH-04 (proposed)	Document the team's actual asynchronous Discord coordination and as-needed synchronous meetings in Section 10.1; add docs/p1/README.md.	P1 instructor C4 feedback asked for meeting frequency and a short repository navigation guide.	Section 10.1 and P1 README only; no changes to existing FR IDs or architecture.	Pending team agreement

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-05 (proposed)	Added a new functional requirement requiring valuable books to receive reinforced box packaging regardless of their weight or size classification. Valuable books are determined either by the bookstore owner or employee.	Direct stakeholder interview identified the need for additional protection when shipping valuable books.	Update FR to prioritize this new one and Table 3.1 to clarify the valuable-book exception; update UC-01 to include the new packaging rule; update the domain model, scenario table, and sequence diagram in Section 5; update the Classification Rules component in Section 6; and add the new requirement to the Section 9 traceability matrix and acceptance tests.	Pending team agreement
CH-06	FR-10 now also saves the book's dimensions and any override reason; acceptance criterion updated.	Review of PR #10 (Patrick): FR-14 shows dimensions and UC-02 saves an override reason, but FR-10 did not say either is saved. The valuable flag is not added because BookPack only reads it from the existing record (FR-16).	None to IDs; section 9 traces FR-10 to the Record Gateway as before. UC-02 postcondition now matches FR-10.	11/08/26

1 to 3 Introduction, Overall Description and Requirements (carried forward).

1. Introduction

1.1 Purpose of this document and of the system

This document establishes the analysis and high-level design baseline for BookPack, a software system that assists bookstore packing employees with book packaging and shipping decisions. It carries forward the P1 requirements and scope, records changes made in P2, and connects the required workflows to the proposed architecture, design decisions, and evidence for P3. Its intended readers are the development team, instructor, and bookstore stakeholders.

BookPack addresses inconsistent manual packaging decisions by recording each book's weight and dimensions, classifying it using the weight and thickness thresholds in Table 3.1, and recommending a cardboard mailer, padded mailer, or reinforced box. A book marked valuable in its existing shipping record receives a reinforced-box recommendation regardless of size class (FR-16, CH-05). Length and width are still captured for the shipping record, but do not determine the classification tier under CH-03.

The system flags heavy books for additional inspection before shipment, supports manual weight entry when a connected scale cannot provide a reading, and updates an existing shipping record with packaging and inspection information. The revised FR-08 acceptance criterion distinguishes manual entry from scale readings near 5.0 lb: a manually entered 5.0 lb must not be flagged and 5.1 lb must be flagged; a scale reading above 5.05 lb must be flagged, while readings within the scale's ±0.05 lb tolerance around 5.0 lb may be flagged (CH-01). The revised validation rule FR-09 is a MUST requirement (CH-02).

These changes reflect simulated scale evidence from spike S-01, not verification with the bookstore's actual scale. The actual scale protocol (Q3) remains open. The bookstore owner approved the Table 3.1 weight/thickness classification thresholds by email on November 5, 2026 (Stakeholder_Interview.md, PR #13); the 5.0 lb inspection flag limit remains unconfirmed (Q1).

1.2 Scope

• In scope: Read weight from a connected scale, with manual entry as a fallback (FR-01, FR-02).

  Out of scope: Manufacture weighing hardware or require a new scale purchase.

• In scope: Record weight in pounds plus manually entered length, width, and thickness in inches (FR-03, CON-02).

  Out of scope: Measure dimensions automatically using cameras or sensors.

• In scope: Classify Small, Medium, or Large using weight and thickness under the bookstore-approved Table 3.1 thresholds (FR-04, CH-03).

  Out of scope: Use length and width as classification thresholds without an approved requirement change.

• In scope: Recommend cardboard mailers, padded mailers, or reinforced boxes for the resulting class (FR-05–FR-07).

Out of scope: Physically pack books or operate packing machines.

• In scope: Retrieve the valuable-book designation from an existing shipping record and recommend reinforced-box packaging regardless of size class (FR-16, CH-05).

Out of scope: Determine a book's monetary value or decide whether it should be designated valuable.

• In scope: Validate weight input and reject values at or below 0 lb and above 70 lb without saving (FR-09, CON-01).

Out of scope: Manage transportation, shipment tracking, or carrier logistics.

• In scope: Apply the revised heavy-book inspection rule and capture inspection outcomes (FR-08, FR-13, FR-14).

Out of scope: Automatically identify physical book damage.

• In scope: Save packaging, weight, and inspection status into an existing shipping record and support review (FR-10, FR-15).

Out of scope: Build sales/payment functions or create missing shipping records (A5).

• In scope: Allow documented packaging overrides when permitted (FR-11, FR-12).

Out of scope: Package multi-book shipments automatically or manage packing-material inventory.

The boundary is the packing-station decision and recording workflow. It assumes an existing station computer and scale (CON-03/A2), an available inspector (A3/Q2), bookstore-approved classification thresholds (A4/Q1 partly answered), and an existing shipping record (A5). Unresolved assumptions are tracked separately rather than presented as confirmed.

1.3 Definitions, acronyms and abbreviations

The P1 definitions remain the baseline, with these P2 clarifications: size class means the Small/Medium/Large tier determined by weight and thickness only (CH-03); flagged means the FR-08 inspection status according to the manual-entry or scale-tolerance acceptance criterion (CH-01); valuable-book designation means a flag retrieved from the existing shipping record that requires reinforced-box packaging regardless of size class (FR-16, CH-05). All other definitions in docs/p1/definitions.md are unchanged since the P1 v0.8 baseline.

1.4 References

[R1] USPS shipping requirement — https://www.usps.com/ship/packages.htm

[R2] Blog explaining book shipping costs — https://goshippo.com/blog/how-to-ship-books-tips-and-best-practices-for-shipping-books

[R3] Student interview transcript — https://docs.google.com/document/d/1TVzl55JsLxMukrQFq5bTf4vavkQTuQM04Xo0r75Nc60/edit?usp=sharing

[R4] Amazon packaging instructions — https://sell.amazon.com/blog/fba-packaging-prep-labeling

[R5] Indeed shipping coordinator description — https://sg.indeed.com/career-advice/finding-a-job/what-does-shipping-coordinator-do

Unchanged since v0.8.

2. Overall Description

2.1 Product perspective and system context

BookPack is a packing-station support system. It receives a book's weight from an external scale or an employee's manual entry, takes employee-entered length, width, and thickness, recommends packaging, requests inspection when the weight warrants it, and saves the result to an existing shipping record. The employee and inspector interact with BookPack directly; the scale and existing shipping-record store remain outside its boundary. Physical weighing, measuring, inspection, packing, delivery, payment, and tracking are not automated by BookPack.

The P1 context diagram (docs/p1/context-diagram.png) describes these external actors and interfaces. Its high-level system boundary is unchanged since P1; P2 architecture diagrams and the interfaces in Sections 5–7 must preserve the same boundary. Under CH-03, only weight and thickness determine the packaging size class; all three dimensions are still recorded.

2.2 Stakeholders and sources

Bookstore manager or owner

Role/need: Reduce inconsistent packaging and unnecessary packing costs; confirm business thresholds.

Evidence/confirmation: P1 literature-based sources; the owner's Nov. 5 interview approved Table 3.1 (Q1) and requested extra protection for valuable books (FR-16). The 5.0 lb inspection limit remains open (Q9).

Packing employee

Role/need: Obtain a usable packaging recommendation, enter measurements, and document any override.

Evidence/confirmation: P1 task analysis and published packaging guidance; validate workflow directly if an employee becomes available.

Inspector or shipping coordinator

Role/need: Resolve flagged-book inspection before shipment proceeds.

Evidence/confirmation: P1 external job-role information; actual availability and separation of duties unconfirmed (Q2/A3).

Student recipient

Role/need: Receive an undamaged book.

Evidence/confirmation: One P1 student interview; this is not a substitute for bookstore requirements confirmation.

Development team and instructor

Role/need: Analyze, implement, and assess a traceable design.

Evidence/confirmation: Course milestones, P1 instructor feedback, version-controlled evidence.

The P1 stakeholder list is otherwise carried forward. In response to instructor feedback, Peter obtained direct owner input on Nov. 5 (Stakeholder_Interview.md, PR #13). Employee workflow, inspector availability and the heavy-book inspection limit still need confirmation.

2.3 User characteristics

Packing employee (primary user): Uses the packing-station computer and scale, enters measurements, follows or overrides a recommendation with an explanation (FR-11/12), and processes books in succession. No specialized shipping-rule knowledge is assumed. Inspector: Records pass or fail for flagged books (FR-13). A flagged book remains pending until the required action is recorded (FR-13). Whether the bookstore has a separate inspector remains open (Q2). These user roles are unchanged since P1 v0.8; CH-01 and CH-03 affect the rules displayed in their workflows, not the identities of the actors.

2.4 Constraints

ID	Constraint	Type	Source
C1	No package recorded by the system may exceed 70 lb, the USPS maximum package weight.	External	USPS link (Section 1.4)
C2	The system needs to store weight in pounds and length, width, thickness in inches.	External	USPS link (Section 1.4)
C3	The system needs to run on the packing station's existing computer system and scale. The bookstore will not buy new hardware.	External	Assumed; not yet confirmed with bookstore (Q3).
C4	All project artifacts must be in a GitHub repository accessible to the instructor.	Organizational	P1 template
C5	Available box sizes for packaging at the store/distributor may be limited.	External	Assumed: limited box sizes; pending bookstore confirmation.

P1 C1 maps to P2 CON-01 (USPS 70 lb limit); C2 to CON-02 (pounds/inches); and C3 to CON-03 (existing hardware, Q3). C4 is an organizational project constraint, not a product requirement. C5 is an unverified packaging-availability assumption; it does not introduce inventory tracking or automated box-stock checks.

The CH-02 upgrade of FR-09 to MUST strengthens validation priority without changing the C1/CON-01 weight limit.

2.5 Assumptions and dependencies

A1 — Book measurements entered or read are accurate.

Consequence/status: Incorrect data leads to an incorrect recommendation; validate inputs and expose review.

A2 — The existing scale can communicate with the computer.

Consequence/status: Manual weight entry becomes the only path; simulated S-01 is encouraging but real hardware is untested (Q3).

A3 — Someone other than the packing employee can inspect flagged books.

Consequence/status: Flagged shipments wait for a qualified person; unresolved (Q2).

A4 — The bookstore agrees to the Table 3.1 weight/thickness thresholds and the separate heavy-book inspection limit.

Consequence/status: Table 3.1 thresholds were approved on Nov. 5 (Q1); the 5.0 lb inspection limit remains unconfirmed (Q9) and may require changes and retesting.

A5 — A shipping record already exists for every book entering packing.

Consequence/status: Additional record creation would exceed the defined scope; not independently verified.

Assumption identifiers remain unchanged since P1. Their evidence and open-question status are updated for P2. Design mitigation for A2 is documented in ADR-02 (direct reading with manual fallback), and S-01 provides simulated evidence only. Open-question status must be changed with actual stakeholder or hardware evidence.

2.6 Feasibility summary

Technical: Core classification, inspection status, and record updates are feasible with familiar application and database techniques; direct scale integration remains conditional on the actual device protocol (Q3). Operational: The owner approved Table 3.1 but does not currently need the system; future adoption, the 5.0 lb inspection limit (Q9), and inspector availability (Q2) remain unconfirmed. Schedule: P2 covers analysis and high-level design; P3 is due December 9, 2026 and requires an agreed implementation/test plan. Resource/economic: The team assumes existing hardware and free development tools; operating-cost savings are plausible goals, not measured results. This assessment carries forward P1's conditional feasibility and incorporates ADR-02/S-01 without claiming the actual scale has been validated.

3. Specific Requirements

Carried forward from P1 v0.8 with CH-01, CH-02, CH-03, CH-05 and CH-06 applied. Existing requirement IDs are unchanged, FR-16 is new, and no requirement has been withdrawn.

3.1 Functional requirements

ID	Requirement (one “shall” sentence)	Priority	Source	Acceptance criterion
FR-01	The system shall read a book's weight from a connected scale.	MUST	Packing employee	Place a known 2.00 lb test weight on the scale; recorded weight is 2.00 lb ± 0.05.
FR-02	The system shall allow manual entry of a book's weight when no scale reading is available.	MUST	Packing employee	With the scale disconnected, enter 1.5 lb; the record stores 1.5 lb.
FR-03	The system shall accept manual entry of a book's length, width, and thickness in inches.	MUST	Packing employee	Enter 9 × 6 × 1; the record stores all three values.
FR-04	The system shall classify each book as Small, Medium, or Large using the thresholds in Table 3.1.	MUST	Bookstore	For 9 test books at and around each threshold, the class matches Table 3.1.
FR-05	The system shall recommend a cardboard mailer for a Small book, unless the book is marked as valuable (see FR-16).	MUST	Bookstore	A Small book not marked as valuable shows “Cardboard mailer.” (Changed, CH-05)
FR-06	The system shall recommend a padded mailer for a Medium book, unless the book is marked as valuable (see FR-16).	MUST	Bookstore	A Medium book not marked as valuable shows “Padded mailer.” (Changed, CH-05)
FR-07	The system shall recommend a reinforced box for a Large book.	MUST	Bookstore, Student	A Large book shows “Reinforced box.”
FR-08	The system shall flag a book for inspection when its weight exceeds 5.0 lb.	MUST	Student	Entered manually: 5.0 lb is not flagged and 5.1 lb is flagged. Read from the scale: any book over 5.05 lb is always flagged; a book within the scale's ±0.05 lb tolerance of 5.0 lb may be flagged. (Changed, CH-01)
FR-09	The system shall reject a weight entry less than or equal to 0 lb or greater than 70 lb.	MUST (was SHOULD, CH-02)	USPS [R1]	Entering 0, -1, or 71 shows an error and saves nothing.
FR-10	The system shall save the weight, dimensions, packaging type, inspection status and any override reason to the book's shipping record.	MUST	Bookstore	After processing, the record shows the weight, length, width, thickness, packaging type and inspection status, and the reason when the recommendation was overridden. (Changed, CH-06)
FR-11	The system shall allow an employee to override a packaging recommendation.	SHOULD	Packing employee	The override is saved, replacing the original recommendation.
FR-12	The system shall require a written reason when an employee overrides a recommendation.	SHOULD	Bookstore	An override with an empty reason is rejected.
FR-13	The system shall allow an inspector to record Pass or Fail for a flagged book.	MUST	Bookstore	A flagged book cannot be marked ready to ship until Pass is recorded.

FR-14	The system shall display all books with an inspection status of “awaiting inspection”, including each book's weight, dimensions, and packaging type.	SHOULD	Inspector	With two flagged books and one unflagged book in the system, the list shows exactly the two flagged books, each with weight, dimensions, and packaging type.
FR-15	The system shall allow an employee to view a book's shipping record, including its weight, packaging type, and inspection status.	SHOULD	Packing employee	After a book is processed, opening its record shows weight, packaging type, and inspection status.
FR-16	The system shall retrieve the valuable flag from a book’s shipping record and, when the flag is set, apply the packaging exception by recommending a reinforced box regardless of the book’s weight or size.	MUST	Bookstore owner (interview, 11/05/2026)	A 0.5 lb, 0.5 in thick book (Small) and a 2.0 lb book (Medium) with the valuable flag set both show “Reinforced box,” and their weight and dimensions are still recorded. The same books without the flag show “Cardboard mailer” and “Padded mailer.” A shipping record with no valuable flag is treated as not valuable. (Added, CH-05)

Table 3.1 — Packaging classification thresholds (approved by the bookstore owner, interview 11/05/2026)

Class	Rule
Small	1.0 lb or less and 1 in thick or less
Medium	3.0 lb or less (and not Small)
Large	More than 3.0 lb

Classification uses weight and thickness only. Length and width are recorded (FR-03) for the shipping record and the inspector's list (FR-14) but do not affect the class (CH-03). A book whose shipping record has the valuable flag set is still measured and classified, but receives a reinforced box regardless of its class (FR-16, CH-05). The bookstore owner approved these thresholds on 11/05/2026 (Stakeholder_Interview); the 5.0 lb flag limit is still open (Q9).

3.2 Quality requirements

Unchanged since v0.8.

ID	Requirement (with threshold, condition and method)	Priority	Source	How it will be verified
QR-01	Ninety-five percent of packaging recommendations shall be returned within 3 seconds of a book's weight and dimensions being entered, measured over one working week under normal load of up to 20 concurrent employees.	MUST	Warehouse operations / instructor	Timestamp logging of request and response, analyzed over a one-week trial
QR-02	The system shall recommend the correct packaging type (mailer, padded mailer, or reinforced box) for at least 98% of books in a 200-book test set, as judged against the documented weight and dimension rules.	MUST	Shipping staff	Run the test set through the system and compare results against the expected packaging table
QR-03	The system shall be available 99% of the time during business hours (8am	SHOULD	Stakeholders	Uptime monitoring log reviewed at the end of the

	to 6pm, Monday to Friday), measured over a 30-day period.			period
QR-04	A new employee shall be able to record a book's weight, view the packaging recommendation, and view the inspection flag within 5 minutes of training, with at least 90% of 10 test users succeeding on the first attempt.	SHOULD	Employees / instructors	Moderated usability test with 10 participants using a task checklist

3.3 Constraint requirements

Unchanged since v0.8.

ID	Constraint (one sentence)	Source	Why it cannot be negotiated
CON-01	The system shall not save a book record if its weight exceeds 70 lb.	R1 USPS package weight	USPS sets limits and will refuse heavier packages. Neither the team nor the bookstore can work around this requirement.
CON-02	The system shall store book weight in pounds and length, width, and thickness in inches.	R1 USPS shipping requirements	USPS publishes limits in pounds and inches. Using other units would conflict at checking points and lead to conversion errors.
CON-03	The system shall run on the packing station's existing computer system and scale without requiring new hardware purchases.	Bookstore (2.4 C3); assumed, Q3	We assumed the bookstore will not buy new hardware. If the answer to Q3 differs, the constraint will be reviewed through change control.

3.4 Initial use cases

The use-case diagram is unchanged since v0.8. Use cases at actor-goal level are listed below; their turn-by-turn expansions are in section 4.2. UC-01 step 6 now reads “the system classifies the book using the weight and thickness thresholds in Table 3.1” (CH-03).

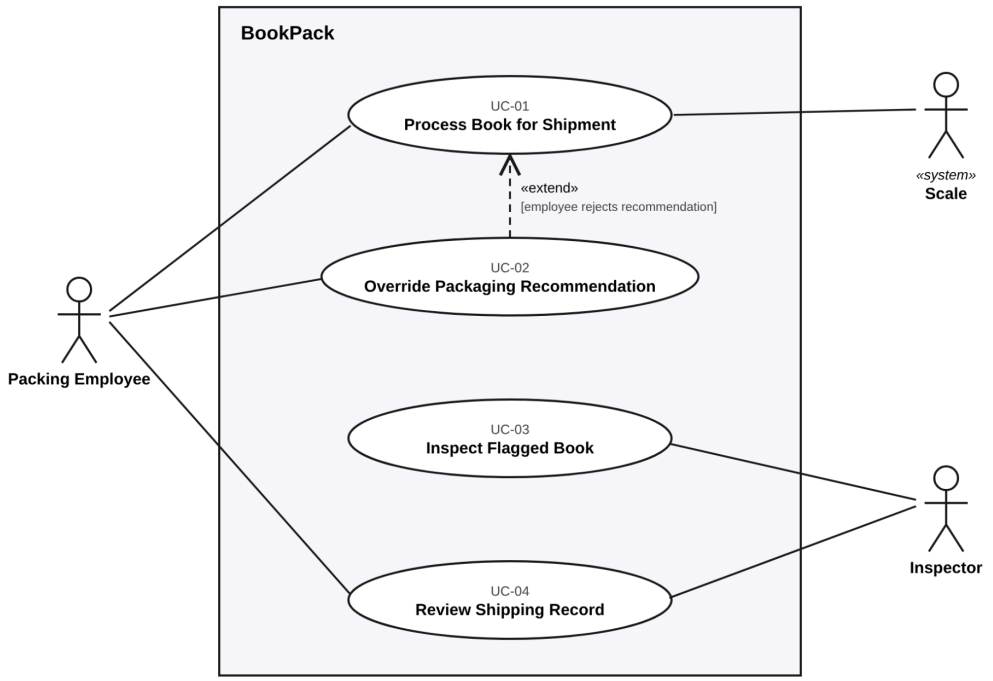


Figure 3.4 — Use-case diagram (unchanged since v0.8)

ID	Use case	Primary actor	Goal
UC-01	Process Book for Shipment	Packing Employee	Get a packaging recommendation and record the book's weight, packaging and inspection status
UC-02	Override Packaging Recommendation	Packing Employee	Replace a recommendation with another packaging type and a written reason
UC-03	Inspect Flagged Book	Inspector	Record Pass or Fail for a book flagged for inspection
UC-04	Review Shipping Record	Packing Employee	View a book's weight, packaging type and inspection status

3.5 Requirements to use-case traceability

Replaced by the extended matrix in section 9.1.

4 Use-Case Analysis

Chapter 7, first half. Here you take the use cases from section 3.4 and write the most important of them out turn by turn, still with the technology switched off.

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Process Book for Shipment	It's the core of the product (FR-01 to FR-10 and FR-16) and most quality requirements (QR-01,QR-02, QR-04); the only use case that works with two external systems(the scale and the shipping record system), each with a failure the employee can see.	(2), (4), (6), (8). (10)
UC-02	Override Packaging Recommendation	The only path where the employee disagrees with the system; carries FR-11 and FR-12, the only rule that rejects an entry for being empty; interacts with the valuable-book exception (FR-16)	(2), (6)
UC-03	Inspect Flagged Book	Only use case with a second actor (Inspector); consumes the FR-08 flag; its failure path is still open (Q4)	(2), (6)

4.2 Expanded use cases

Copy the block below once for each use case in 4.1.

UC-01 Process Book for Shipment

Primary actor	Packing Employee
Supporting actors and external systems	Scale(external device); existing shipping record system(external system)
TUCBW (this use case begins when)	The packing employee starts processing the book for shipment by giving its order number
TUCEW (this use case ends when)	The packing employee sees confirmation that the book's weight, packaging type and inspection status were saved to its record.
Preconditions	The book has to have been picked for an order and is at the packing station.
Postconditions	The shipping holds the book’s weight, length, width, thickness, packaging type and inspection status. A book over 5.0 lbs is awaiting inspection and can’t be marked ready to ship until Pass is recorded for it (UC-03).
Traces to requirements	FR-01 to FR-10, FR-16; QR-01, QR-02, QR-04;, CON-01, CON-02

Actor input / action	System response
	(0) The system shows that it's ready to process a book and asks for order number
(1) The packing employee gives the book order number	(2) The system finds the book's shipping record, shows the order number and title, and asks the employee to place book on scale.
(3) The packing employee places book on scale.	(4) The system reads the weight and shows it, and asks for the book length, width and thickness.
(5) The packing employee enters the length, width, and thickness in INCHES.	(6) The system classifies the book using the weight and thickness thresholds that are set in Table 3.1 and shows its size category with weight and thickness.
(7) The packing employee asks for packaging type recommendation. .	(8) The system shows the recommended packaging type according to Table 3.1 and displays if the book requires additional inspection under FR-08.
(9) The packing employee accepts the recommendation.	(10) The system saves the weight, dimensions, packing type and inspection status to the shipping record and confirms that the book was processes.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(2)	No shipping record matches the order number	The system says no record found and saves nothing	Rejoins (1)
A2	(3)	No scale reading is available	The packing employee enters the weight by hand (FR-02)	Rejoins (4) with the entered weight
A3	(4)	The weight is 0lbs or less, or over 70lbs	The system shows error and saves nothing (FR-09, CON-01)	Rejoins (3)
A4	(8)	The book's weight meets the FR-08 inspection criteria, accounting for the applicable manual entry or scale-reading tolerance	The system shows the recommendation and that the book requires additional inspection(FR-08). At step (10) the inspection status is saved as awaiting inspection.	Rejoins (9)
A5	(8)	The shipping record marks the book as valuable	The system recommends a reinforced without regards to size class and says why; the weight and dimensions are still recorded (FR-16)	Rejoins (9)

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A6	(9)	The packing employee rejects the recommendation	The employee chooses another packing type with a reason	Continues in UC-02 Override Packing Recommendation
A7	(10)	The shipping record cannot be saved	The system says the record was not saved and keeps the entries on screen.	Rejoins (9)

UC-02 Override Packaging Recommendation

Primary actor	Packing Employee
Supporting actors and external systems	Existing shipping record system (external system)
TUCBW (this use case begins when)	The packing employee rejects the packaging BookPack recommended for the book being processed (UC-01 alternate flow 9a)
TUCEW (this use case ends when)	The packing employee sees confirmation that the chosen packaging and the reason were saved to the book's shipping record
Preconditions	The employee is processing a book in UC-01 and BookPack has shown a recommended packaging type for it.
Postconditions	The shipping record holds the packaging type the employee chose, in place of the recommendation, together with the written reason. The book's inspection status is unchanged.
Traces to requirements	FR-11, FR-12, FR-10; FR-05 to FR-07 and FR-16 (the recommendation being overridden)

Actor input / action	System response
	(0) The system shows the recommended packaging type with the book's size class, weight and thickness, and offers the choice to use a different packaging type.
(1) The packing employee asks to use a different packaging type.	(2) The system shows the other packaging types and asks for a reason.
(3) The packing employee chooses a packaging type and writes the reason.	(4) The system shows the chosen packaging type and the reason and asks the employee to confirm.
(5) The packing employee confirms.	(6) The system saves the chosen packaging type and the reason to the shipping record in place of the recommendation and confirms that the override was saved.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(4)	The reason is empty	The system says a reason is required and saves nothing (FR-12)	Rejoins (3)
A2	(3)	The employee chooses the packaging type that was already recommended	The system says this matches the recommendation and no override is needed	Ends; UC-01 continues at step 10 with the recommendation
A3	(5)	The employee cancels	The system keeps the original recommendation and saves nothing	Ends; UC-01 continues at step 9
A4	(6)	The override cannot be saved to the shipping record	The system says the override was not saved and keeps the chosen type and reason on screen	Rejoins (5)

UC-03 Inspect Flagged Book

Primary actor	Inspector
Supporting actors and external systems	Existing shipping record system (external system)
TUCBW (this use case begins when)	The inspector asks for the books awaiting inspection
TUCEW (this use case ends when)	The inspector sees confirmation that the result was saved and the book is no longer on the awaiting list
Preconditions	The user is acting as the inspector (A3, Q2).
Postconditions	The result is in the shipping record. A passed book is ready to ship; a failed book stays blocked and returns for repackaging (A2). In both cases the book leaves the awaiting list.
Traces to requirements	FR-13, FR-14, FR-10; consumes FR-08

Actor input / action	System response
	(0) The system shows how many books are awaiting inspection.
(1) The inspector asks to see the books awaiting inspection.	(2) The system lists each flagged book with order number, title, weight, length, width, thickness and packaging type.
(3) The inspector selects a book.	(4) The system shows that book's details and asks for Pass or Fail.
(5) The inspector records Pass.	(6) The system saves Pass, marks the book ready to ship, removes it from the list and confirms.

The physical check of the book happens between (4) and (5), outside the system.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(2)	No book is awaiting inspection	The system says no books are awaiting inspection	Ends
A2	(5)	The inspector records Fail	The system saves Fail, keeps the book blocked and marks it for repackaging (assumed, Q4)	Ends after confirmation
A3	(6)	The result cannot be saved	The system says the result was not saved; the book stays on the list	Rejoins (5)

4.3 Self-review log

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-03	Precondition or alternate flow	P1 precondition “at least one book awaiting inspection” can be false during normal work	Moved to A1 (empty list)
UC-03	No background processing	P1 step 4 (physical check) was not a system turn	Removed from the table; noted outside it
UC-03	Step (0) present	P1 had no step (0)	Added: count of books awaiting inspection
UC-03	Acknowledgment at end	P1 ended with confirmation; kept	No defect
UC-02	Step (0) present	UC-02 had no expansion in P1, so no step (0) existed	Added (0): the recommendation is shown with class, weight and thickness so the employee can judge it
UC-02	Precondition or alternate flow	“The employee gives a reason” can be false during normal work	Kept out of the preconditions; handled as A1 (FR-12)
UC-02	Real alternate flows	FR-11 does not say what happens if the employee picks the type that was already recommended	Added A2 (no override recorded)
UC-02	Acknowledgment at end	None; (6) ends with a confirmation the employee can see	No defect
UC-02	No technology	None; no screens, widgets or storage named	No defect

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-02	Traces to requirements	Whether an employee may override the reinforced box a valuable book receives (FR-16) is not specified	Not invented; raised as open question Q8
UC-01	Step(0) present	P1 had no step (0)	Added (0): the system shows it's ready and asks for the order number
UC-01	Precondition or alternate flow	P1 precondition "an existing shipping record" can be false during normal work(wrong order number)	Moved to A1
UC-01	One turn per row	P1 steps 6, 7 and 8 were three system steps in a row with no actor to turn between them	Split into (6) classification, (7) employee asks for recommendation, (8) recommendation and inspection check
UC-01	No technology	P1 step 3 said the weight "displays on the user interface"	Reworded: the system reads the weight from the scale and shows it.
UC-01	Acknowledgement at the end	P1 step 10 updated the record but gave the employee no confirmation	(10) now ends with a confirmation that book was processed
UC-01	Traces to requirements	P1 step 6 used "dimension thresholds", not table 3.1 (CH-03), and UC-01 had no path for valuable books (CH-05)	Step(6) uses the weight and thickness thresholds in Table 3.1; added A5 tracing to FR-16
UC-01	Real alternate flows	P1 had no flow for a record that can't be saved	Added A7

5 Analysis Models

Chapter 7, second half. What has to exist inside the system for the use cases in section 4 to happen, and how those parts talk to each other.

5.1 Domain model

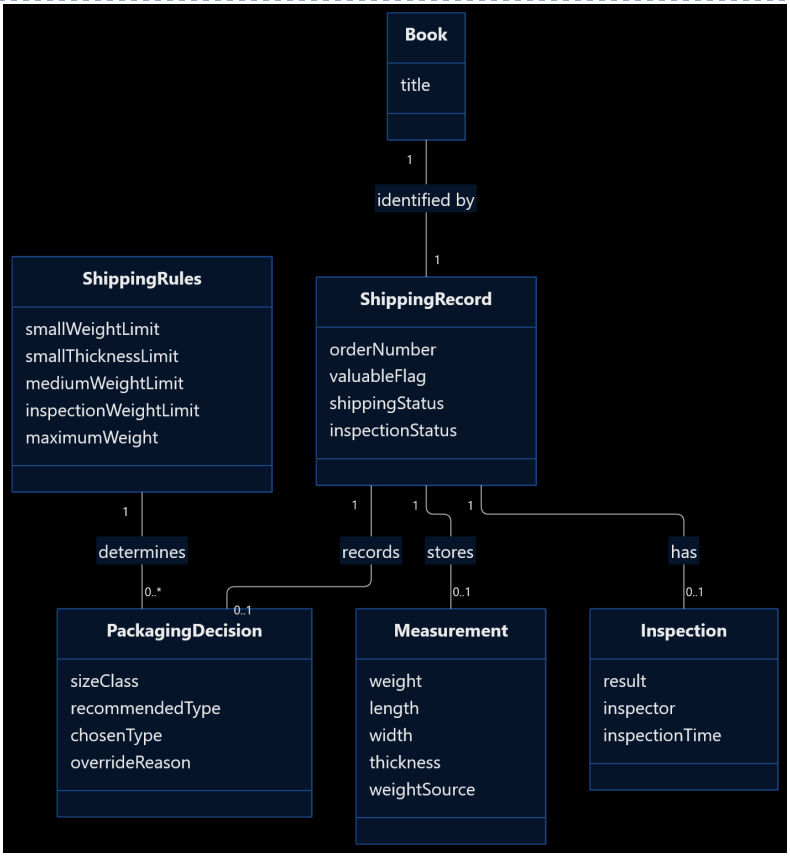


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
Book	Physical book being shipped	title	Book 1 - 1 ShippingRecord
Measurement	Recorded book measurements	weight, length, width, thickness, weightSource	Measurement 0..1 - 1 ShippingRecord
Shipping Record	Existing record tracking a book's shipment	shipment orderNumber, valuableFlag, shippingStatus, inspectionStatus	ShippingRecord 1 - 1 Book; 1 - 0..1 Measurement; 1 - 0..1 PackagingDecision; 1 - 0..1 Inspection
Packaging Decision	Classification and packaging selection	sizeClass, recommendedType, chosenType, overrideReason	PackagingDecision 0..1 - 1 ShippingRecord; PackagingDecision 0..* - 1 ShippingRules

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
Shipping Rules	Business rules determining packaging classification and inspection requirements	smallWeightLimit, smallThicknessLimit, mediumWeightLimit, inspectionWeightLimit, maximumWeight	ShippingRules 1 - 0..* PackagingDecision
Inspection	Inspection result for a flagged book	result, inspector, inspectionTime	Inspection 0..1 - 1 ShippingRecord

5.2 Object interaction models

5.2.1 UC-02 Override Packaging Recommendation

(a) Non-trivial steps: (2) retrieves available packaging types; (6) saves the chosen type and reason. Step (4)'s empty-reason validation is handled in alternate flow A1 (FR-12).

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Packing Employee	asks to use a different packaging type		PackingController
2	PackingController	retrieves available packaging types	packaging types	PackagingDecision
3	PackingController	shows the other packaging types and asks for a reason	packaging types	Packing Employee
4	Packing Employee	chooses a packaging type and writes the reason	chosen type, reason	PackingController
5	PackingController	shows the chosen packaging type and the reason and asks to confirm	chosen type, reason	Packing Employee
6	Packing Employee	confirms	confirmation	PackingController
7	PackingController	saves the chosen packaging type and reason to the shipping record	chosen type, reason	ShippingRecord
8	ShippingRecord	confirms save successful		PackingController
9	PackingController	confirms that the override was save	confirmation	Packing Employee

(c) Analysis sequence diagram.

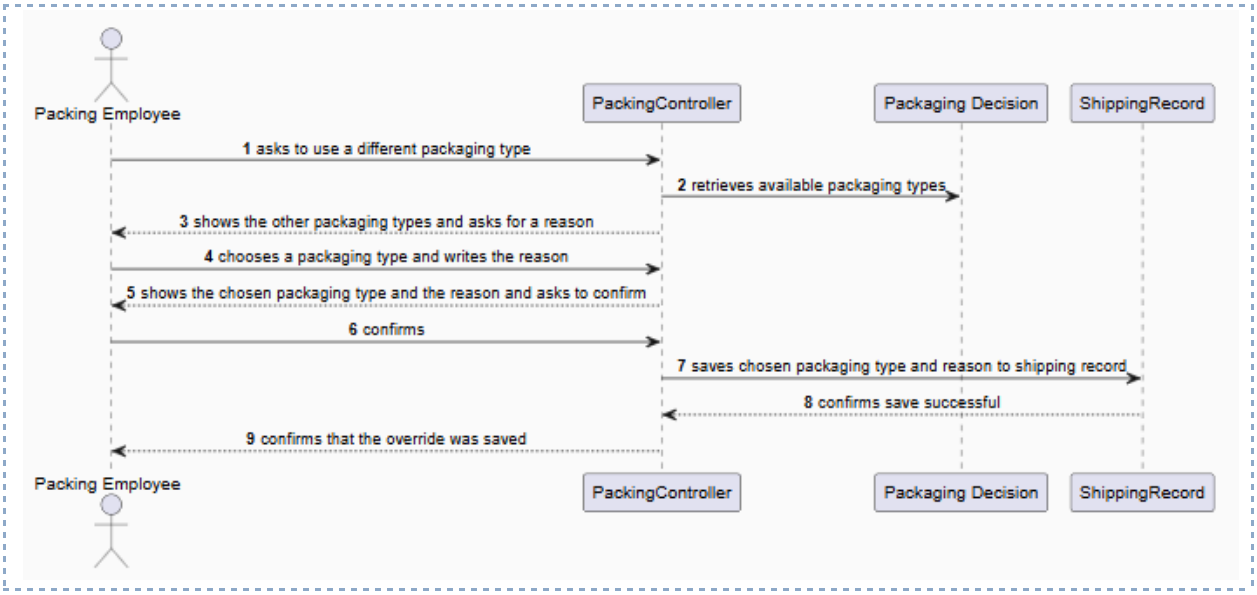


Figure 5.2.1 Analysis sequence diagram for UC-02; message numbers match the scenario table. PackingController is an interaction object that controls UC-02

5.2.2 UC-03 Inspect Flagged Book (Pass scenario)

(a) Non-trivial steps. (2) finds the records awaiting inspection; (6) records the result and changes the record's status. (0) shows a count; (4) shows details already listed; (1), (3), (5) are actor inputs.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Inspector	asks for the books awaiting inspection	—	InspectionController
2	InspectionController	asks for the records awaiting inspection	status	ShippingRecord
3	InspectionController	shows each book with weight, dimensions and packaging type	Measurement, PackagingDecision	Inspector
4	Inspector	selects a book	order number	InspectionController
5	InspectionController	asks for the book's open inspection	order number	Inspection
6	Inspector	records Pass	result	InspectionController
7	InspectionController	records the result, inspector and time	Pass, Inspector	Inspection
8	InspectionController	sets the record to ready to ship	status	ShippingRecord
9	InspectionController	confirms the result and shows the updated list	—	Inspector

(c) Analysis sequence diagram.

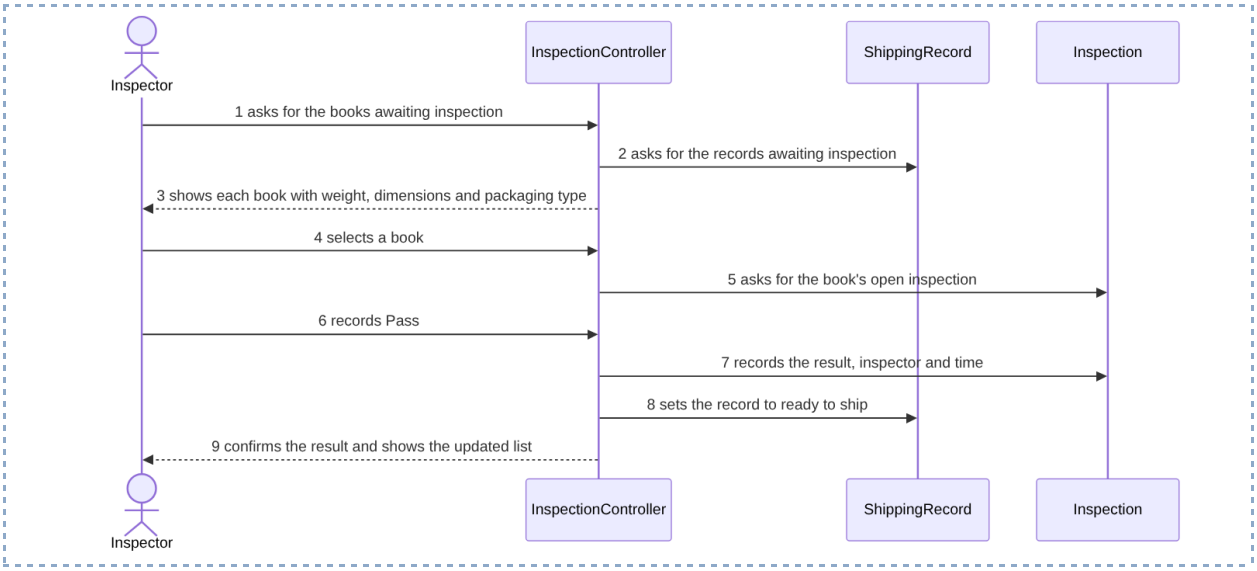


Figure 5.2.2 Analysis sequence diagram for UC-03, Pass scenario; message numbers match the scenario table. InspectionController is an interaction object that controls UC-03.

5.3 Consistency cross-check

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	Pass	n/a
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Pass	n/a
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	**Failed for UC-02.** Fig. 5.2.1 has 9 messages but the 5.2.1(b) table had 7 rows. Messages 2 and 8 had no rows, and later numbers were shifted. UC-03 passes (9 rows, 9 messages).	5.2.1(b) rebuilt with 9 rows
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	**Failed for UC-02.** Step (4) was listed as non-trivial, but no row validates anything (the empty-reason check is A1). 4.1 also disagreed with 5.2.1(a). UC-03 passes.	5.2.1(a) revised to match 4.1.
K5 Every analyzed use case traces to at least one functional requirement	Pass	n/a

Check	Result (pass, or what failed)	Fixed in
K6 The last messages of each diagram make the use case postcondition true	Pass. UC-02 messages 7–9 save the chosen type and reason, then confirm. UC-03 messages 7–9 record Pass, mark the book ready to ship, and show the updated list. The Fail scenario is handled separately in alternate flow A2.	n/a

6 High-Level Design (Architecture)

Chapter 8. The rule now reverses: a high-level design that does not say where things run, what they are built from and what that choice cost has not been done.

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Performance	An employee enters the last dimension while up to 20 stations are in use	Packing Workflow and Classification Rules	Shows size class, recommendation and flag	95% within 3 s over one working week; scale read uses at most 1.05 s of that (S-01 p95)	QR-01
QAS-02	Availability	The scale stops sending or is unplugged during weighing	Scale Reader	Detects the loss and switches to manual entry with no work lost	Detected within 1.5 s; system usable 99% of business hours over 30 days	QR-03
QAS-03	Correctness	A 200-book test set, including books on every Table 3.1 and 70 lb boundary and books marked valuable, is processed	Classification Rules	Returns the class, packaging type and flag	At least 98% match the expected table; 100% of books over 70 lb rejected	QR-02
QAS-04	Modifiability	The bookstore changes a Table 3.1 threshold after answering Q1	Rules configuration	New threshold takes effect with no code change	Changed and boundary-retested in under 1 hour, touching one file	QR-02, A4

Constraints that shape the architecture: CON-03 (existing PC and scale) puts all BookPack components on the station PC with no new server; CON-01 and CON-02 put the 70 lb rule and units in one Classification Rules component so they are checked in one place.

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Layered (presentation, application, rules, integration) with client-server access to the shipping record system	QAS-01 (in-process calls, no network hops except the final save); QAS-04 (rules isolated in one layer); QAS-02 (scale isolated in the integration layer)	Some pass-through calls between layers; one deployment per station to update	Chosen

Style considered	Drivers it serves	What it costs	Verdict
Microservices (scale, rules, records as separate services)	Independent deployment of each part	Network calls inside a 3 s budget; a server to run them, against CON-03; far more operations work than 6 small components need	Rejected
Pipe-and-filter (weight → validate → classify → flag → save)	Matches the straight-line part of UC-01	Override (UC-02), manual fallback and inspection (UC-03) loop back or wait for a person, which a one-way pipeline does not handle	Rejected

We chose a layered design because every BookPack use case is a short request the employee waits for, so the 3 s budget in QAS-01 favors in-process calls over network hops. Isolating the classification rules in their own layer serves QAS-03 and QAS-04: rules are tested and changed in one place. Isolating the scale and the record system in an integration layer serves QAS-02 and contains the hardware risk from Q3. ADR-01 records this decision.

6.3 Component view

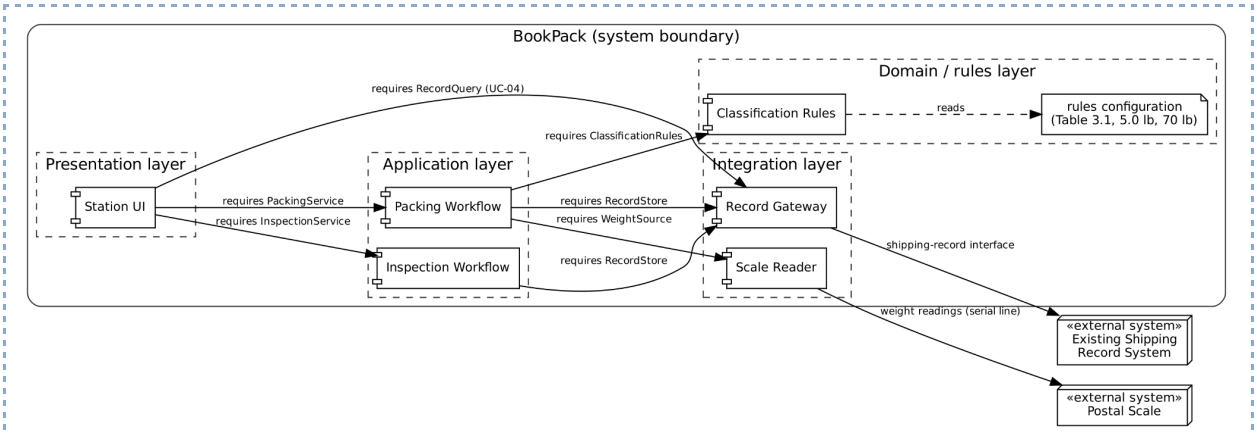


Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Station UI	Shows the packing and inspection workflows and collects input	— (actor-facing)	PackingService, InspectionService, RecordQuery	UC-01 to UC-04, QR-04
Packing Workflow	Runs UC-01 and UC-02 from weighing to saving	PackingService	WeightSource, ClassificationRules, RecordStore	UC-01, UC-02, QAS-01
Inspection Workflow	Runs UC-03: lists flagged books and records Pass or Fail	Inspection Service	RecordStore	UC-03, FR-13, FR-14

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Classification Rules	Decides size class, packaging type, flag and weight validity from the rules configuration, applying the valuable-book exception first; the 70 lb limit is fixed in code	ClassificationRules	Rules configuration file	QAS-03, QAS-04, ADR-03, CON-01, FR-16
Scale Reader	Returns a stable weight or reports that the scale is unavailable	WeightSource	Postal scale serial line	QAS-02, ADR-02, S-01
Record Gateway	Reads and updates the bookstore's existing shipping records, including the valuable flag	RecordStore, RecordQuery	Shipping record system interface	FR-10, FR-15, FR-16, ADR-01

External systems outside the boundary: Postal Scale and the Existing Shipping Record System.

6.4 Deployment view

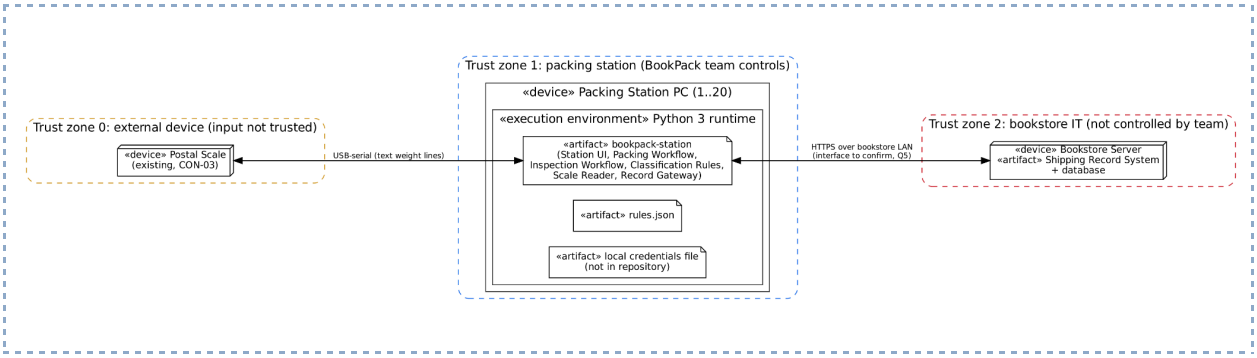


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Packing Station PC (1 to 20), Python 3 runtime	bookpack-station artifact (all six components), rules.json, local credentials file kept out of the repository	Postal scale (USB-serial, text lines); Bookstore server (HTTPS over the store LAN)	Zone 1: controlled by the team
Postal Scale (existing)	Scale firmware	Station PC (USB-serial)	Zone 0: external device; every reading is validated (ADR-02, FR-09)
Bookstore Server (existing)	Shipping record system and its database	Station PCs (HTTPS; exact interface open, Q5)	Zone 2: bookstore IT, not controlled by the team

Trust boundaries: scale → station (untrusted device input, filtered by Scale Reader) and station → bookstore server (authenticated with a station account; credentials stay in local configuration). No cloud service is used, so there is no shared-responsibility split. The inspector uses a station PC in the inspector role (Q6).

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
PackingController	Packing Workflow	—
InspectionController	Inspection Workflow	—
ScaleReader	Scale Reader	Wraps the external scale
ShippingRules, SizeClass, PackagingType	Classification Rules	Merged: all are rule data loaded from rules.json
Measurement, PackagingDecision, Inspection	Packing Workflow (created); Inspection Workflow updates Inspection	Split: created in the workflows, stored through Record Gateway
ShippingRecord, Book	Record Gateway	External: owned by the bookstore's shipping record system
PackingEmployee, Inspector	Station UI	Actors; identity comes from the station login

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	Rules, scale access and record access are separate components	6.3; ADR-01
Information hiding	The scale's line format is known only to Scale Reader; the record system's interface only to Record Gateway	WeightSource, RecordStore; ADR-02
Low coupling	Workflows depend on interfaces, not on the scale or record system	6.3 Requires column
High cohesion	Classification Rules holds every threshold and nothing else	ADR-03
Design for change	Thresholds live in rules.json because Q1 may change them; the scale parser can be swapped when Q3 is answered	QAS-04; ADR-02, ADR-03
Keep it simple (KISS)	One process per station, no server or broker added	ADR-01; CON-03

7 Design Rationale

7.1 Architecture decision records

At least three. ADR-01 records the style decision from 6.2. Use the format you practiced in the Lesson 16 workshop, written about your own project. Copy the block for each decision.

ADR-01 Use a layered architecture on each packing station

Status	Accepted
Date and owner	2026-11-08, Ahmet Keles
Context (drivers)	QAS-01 (3 s for 95%), QAS-02 (scale loss), QAS-04 (threshold change); CON-03 (existing PC, no new hardware); UC-01 to UC-03 are synchronous requests the actor waits for
Options considered	Layered application on each station PC, with client-server access to the existing shipping record system
Options considered	Microservices for scale, rules and records on a separate server
Options considered	Pipe-and-filter pipeline from weighing to saving
Decision	Option 1, layered (presentation, application, rules, integration)
Consequences	Favors performance (in-process calls) and modifiability (rules in one layer). Costs: each of up to 20 stations must be updated when the code changes; layers add pass-through calls; one station's failure is not covered by another
Evidence	Spike S-01 shows the scale read uses 1.05 s at p95, leaving about 1.95 s for classification and saving in one process
Related	Section 6.2, QAS-01, QAS-02, QAS-04, CON-03

ADR-02 Read the scale directly, with manual entry as the fallback

Status	Accepted
Date and owner	2026-11-07, Nicholas Suther
Context (drivers)	FR-01 and FR-02; QR-01 (recommendation within 3 s); CON-03 (existing computer and scale, no new hardware); A2 (scale can connect to the computer, unconfirmed, Q3); risk R1; UC-01 steps 2 to 4 and alternate flow 3a.
Options considered	Direct read: BookPack reads the scale's data stream, waits for a stable reading, and switches to manual entry if the scale is silent.
Options considered	Keyboard wedge: the scale types its weight into whichever field has focus, so no scale code is needed.
Options considered	Manual entry only: the employee reads the scale display and types the weight.

Decision	Option 1, direct read with manual fallback.
Consequences	Favors accuracy and usability: no typing errors, and only a stable weight is accepted, which a keyboard wedge cannot tell apart from a settling one; a wedge can also type into the wrong field. Costs: code tied to the scale's data format that must change if the scale model changes; about 1 s before a disconnected scale is noticed; scale noise of about ±0.02 lb makes books at exactly 5.0 lb unreliable, so FR-08's acceptance criterion changed.
Evidence	Spike S-01 (section 8): 105/105 reads within ±0.05 lb, p95 1.05 s, and 20/20 disconnects fell back to manual entry within 1.22 s, against a simulated scale. Not yet tested on the bookstore's actual scale (Q3).
Related	R1, QR-01, CON-03, A2, Q3, S-01, CH-01

ADR-03 Keep classification thresholds in a configuration file

Status	Accepted
Date and owner	2026-11-08, Ahmet Keles
Context (drivers)	Table 3.1 thresholds were approved by the owner on 11/05 but may still change, and the 5.0 lb flag limit is still open (Q1, A4, R5); QAS-03, QAS-04; CH-01 added a scale tolerance
Options considered	Hard-code the thresholds in the classification code
Options considered	Store them in rules.json on each station, loaded and validated at start-up
Options considered	Store them in the bookstore's shipping record database
Decision	Option 2, rules.json read by Classification Rules
Consequences	Favors modifiability: a Q1 answer changes one file with no code change. Costs: the file must be validated at start-up (a typo could misclassify every book); 20 stations need the same file; the 70 lb CON-01 limit and the FR-16 valuable-book exception stay in code so a config edit cannot weaken them
Evidence	None yet; boundary tests in P3 will load two different rule files and check both
Related	QAS-04, Q1, A4, R5, CON-01, CH-01, CH-05, FR-16

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
Accuracy and usability of weight capture	Read the scale directly and accept only stable readings, with manual fallback	Scale-specific reader code; ~1 s disconnect detection; FR-08 boundary needs a tolerance rule	ADR-02

Quality favored	Architectural move	Cost accepted	Recorded in
Performance and simplicity	Layered app running in one process on each station	Updates on up to 20 stations; no failover between stations	ADR-01
Modifiability of rules	Thresholds in rules.json, 70 lb limit kept in code	Start-up validation needed; file must match on every station	ADR-03

8 Prototype Evidence: Architecture Spike

Spike ID and title	S-01 Reading the packing-station scale
Risk or decision it addresses	R1, ADR-02, QR-01 (FR-01, FR-02, A2)
Question (answerable yes or no)	Can BookPack read a stable weight from a serial/USB postal scale within ±0.05 lb (FR-01), within QR-01's 3 s budget, and detect a missing scale fast enough to switch to manual entry (FR-02)?
Set-up	Python 3 standard library. The bookstore's scale is not available yet (Q3), so a simulated serial postal scale runs on a pseudo-terminal: about 10 readings/s, unstable readings while settling, ±0.02 lb noise, 5% garbled lines, and "unplugged" cases. The reader waits for two agreeing stable readings and falls back to manual entry after 1.0 s of silence or 3.0 s without a stable reading. Folder: /spikes/S-01-scale-read/
How it was measured	105 normal trials (15 each at 0.50, 1.00, 2.00, 3.00, 5.00, 5.10 and 12.00 lb, on the Table 3.1 and FR-08 boundaries) and 20 disconnect trials; each trial logged the weight returned, the error and the elapsed time to results.csv.
Result (numbers)	105/105 reads within ±0.05 lb (max error 0.020 lb). Time to stable weight: median 0.73 s, p95 1.05 s, max 1.25 s. 30 of 812 garbled lines ignored with no wrong reading. 20/20 disconnects fell back to manual entry, detected in at most 1.22 s. A 5.00 lb book read as over 5.0 lb in 5 of 15 trials.
What it changed or confirmed	Confirmed ADR-02: reading uses about 35% of QR-01's 3 s at p95. Changed FR-08's acceptance criterion (CH-01) because scale noise makes the exact 5.0 lb boundary unreliable. Q3/A2 is still open: the real scale has not been tested.
Where to see it	/spikes/S-01-scale-read/ (README.md, results.csv, results.md), commit ce38eaf

9 Traceability

9.1 Extended traceability matrix

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	UC-01 (3)-(4)	Measurement	Scale Reader, Packing Workflow	QAS-02, ADR-02	S-01 (105/105 reads)
FR-02	UC-01 A2	Measurement	Station UI, Packing Workflow	QAS-02, ADR-02	S-01 (20/20 fallbacks)
FR-03, CON-02	UC-01 (5)	Measurement	Station UI, Packing Workflow, Record Gateway	none	Entry and unit tests
FR-04 to FR-07, QR-02	UC-01 (6), (8), A5	ShippingRules, PackagingDecision	Classification Rules	QAS-03/04 ADR-01/03	200-book test set, planned
FR-08	UC-01 A4; UC-03	ShippingRules, Inspection	Classification Rules, Packing Workflow	QAS-03, ADR-02/03	5.0/5.1 ln test, planned; S-01
FR-09, CON-01	UC-01 (4), A3	ShippingRules, ScaleReader	Classification Rules, Scale Reader	QAS-03, ADR-02/03	0/-1/71 lb tests
FR-10	UC-01 (10); UC-02 (6), UC-03 (6)	Measurement, PackagingDecision, ShippingRecord, Inspection	Packing/Inspection Workflow, Record Gateway	ADR-01	Record field test
FR-11, FR-12	UC-02 (1)–(3), A1	PackagingDecision, PackingController	Packing Workflow, Station UI	none	Override tests
FR-13, FR-14	UC-03 (1)-(2), (5)-(6)	InspectionController, Inspection	InspectionWorkflow, Record Gateway	none	Pass and list tests
FR-15	UC-04 (not analyzed)	Shipping Record	Record Gateway,Station UI	ADR-01	Record view test
FR-16	UC-01 A5	ShippingRecord, PackagingDecision, Shipping Rules	Record Gateway, Packing Workflow, Classification Rules	ADR-01/03	Valuable-book test
QR-01	UC-01 (5)–(8)	PackingController, ShippingRules	Packing Workflow, Classification Rules	QAS-01, ADR-01/02	S-01 (p95 1.05s); trial
QR-03	UC-01 (3)–(4), A2	ScaleReader	Scale Reader	QAS-02, ADR-01/02	S-01 (1.22s); uptime log
QR-04	UC-01 (0)–(10)	PackingController	Station UI	none	Usability test
CON-03	UC-01 (3)–(4)	ScaleReader	All six (6.4)	ADR-01/02	S-01

9.2 Gap analysis

Gap	Items	What you did about it
Requirements with no component	None	Every FR, QR, and CON reaches a component in 6.3.
Components with no requirement, QAS or ADR	None	Every 6.3 component answers to a use case, QAS or ADR.
Use cases not analyzed (and why that is acceptable)	UC-04	It only realizes FR-15(SHOULD), a read-only view through Record Gateway
Other gaps	QR-04 has no QAS; QR-01 trial, QR-03 uptime log, QR-04 test and S-01 on the real scale (Q3) not yet run	QR-04 is deferred as SHOULD. The unrun checks move to P3 (10.2)

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

P1's assignment of responsibilities carries into P2 unless team members explicitly agree otherwise. Section 0.1 of the final report must record actual P2 commits, pull requests, and reviews, not merely these expected responsibilities.

• Peter Yu: Team coordination, stakeholder information and references, classification-threshold confirmation, and submission review.

• Patrick Paul: Purpose and scope, project planning and risks, repository documentation, document consistency, and final P2 review.

• Nicholas Suther: Functional requirements, classification and acceptance criteria, and scale integration investigation.

• Jasmine Tran: Quality requirements, measurement validation, traceability, and design review.

• Afeef Zarraf: Constraints and feasibility, use cases, shipping-record integration, and interfaces.

• Ahmet Keles: Context diagrams, assumptions/open questions, inspection flow, and architecture.

• Lucas Pham: Packing-employee and inspector workflows and interface review.

Communication and meetings: The team initially planned to meet once per week, but routine coordination evolved into asynchronous Discord messages and repository/issue discussions. Members raise blockers and review one another's work asynchronously. Synchronous meetings are arranged when needed for unresolved design choices, deadlines, or coordination; a weekly synchronous meeting has not consistently occurred. This reflects actual practice rather than implying meetings were held each week.

The working agreement remains to seek consensus, escalate unresolved decisions to the team, and document changes affecting requirements or architecture. A majority vote can resolve an internal disagreement, but it cannot substitute for bookstore approval of stakeholder-dependent assumptions such as Q1. For P2 contributions, Patrick proposes a branch and pull request per independently reviewable section, a teammate review, and merging only once comments have been addressed. Previous P1 commits went directly to main; use actual PR history, not the proposed process, as evidence in Section 0.1.

10.2 Plan to P3

Milestone	Date	Owner	Done when
Triage P2 feedback and assign changes	Nov. 12 (proposed)	Peter, Patrick, section owners	Each comment logged, assigned and resolved or justified.

Milestone	Date	Owner	Done when
Confirm inspection limit and investigate actual scale	Nov. 18 (proposed)	Peter (stakeholder), Nicholas (scale)	Direct answer on the 5.0 lb inspection limit (Q9), with Q3 resolved or escalated; real-device test if available.
Complete detailed design and interfaces	Nov. 25 (proposed)	Ahmet, Lucas, Afeef; Jasmine (traceability)	P3 interfaces and detailed components trace to P2 FR IDs.
Integrate core workflows	Dec. 2 (proposed)	Implementation owners to confirm	Weight, classification, inspection and record updates work end to end.
Complete verification tests	Dec. 5 (proposed)	Jasmine, Nicholas, implementation owners	Boundary/failure/QR/persistence tests recorded, defects owned.
Freeze and review P3 candidate	Dec. 8 (proposed)	Patrick, Peter, all members	Document, diagrams, code evidence, PRs, contributions and PDF agree.
Submit P3	Dec. 9 (official deadline)	Canvas submitter to confirm	Required artifacts committed/tagged and one Canvas submission completed on time.

The confirmed course deadline for P3 is December 9, 2026. The internal milestones below are a proposed schedule for team review, with owners and completion conditions; they are not reported as commitments already approved by all members.

10.3 Risks (updated)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1	Partially mitigated by simulated S-01; real scale still untested (Q3).	Medium / High	Nicholas: test real protocol and failures; retain ADR-02 manual fallback; log FR-01/02 tests.
R2	Open; spike does not verify application validation or FR-08/09 boundaries.	Medium / High	Jasmine: test negative/zero/70/>70 lb, manual 5.0/5.1 and scale tolerance; rejected data not saved.

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R6 (new)	New ADR-02 risk: direct scale reader depends on actual hardware protocol (Q3).	Medium / High	Nicholas: collect real scale messages, isolate parser and test disconnect/reconnect with fallback.
R3	Open; inspector availability assumption A3 and Q2 remain unconfirmed.	Medium / High	Ahmet: confirm coverage; flagged shipments remain pending until inspected (FR-13/14).
R4	Open; end-to-end persistence and error handling not yet demonstrated.	Medium / High	Afeef: test record updates, failed writes and stored status (FR-10/15; A5).
R5	Closed; received Table 3.1 thresholds bookstore confirmation (Q1/A4).	Medium / High	Peter: obtained direct elicitation; Nicholas/Jasmine updated class rules and tests.
R7 (new)	S-01 showed nominal 5.00 lb readings sometimes crossing 5.0 threshold.	Medium / Medium	Jasmine/Nicholas: verify CH-01 tolerance and stakeholder acceptance (Q1).

P1's risks R1–R5 remain identifiable. Spike S-01 tested a simulated scale, so it reduces uncertainty but does not establish real-device compatibility. The likelihood/impact judgments and new risks R6/R7 are proposals for team confirmation.

No risk is marked closed without evidence. Record status changes in the P2 revision log and link corresponding FR, QR, ADR or spike results where relevant.

10.4 Repository

The public team repository is https://github.com/PeterYuGit/Team7_Repository. P1 sources are in docs/p1/; current P2 source contributions belong in docs/p2/, including design decisions and section files; the source for P2 diagrams should be placed in docs/p2/diagrams/; prototypes and test evidence belong under spikes/ (currently spikes/S-01-scale-read/).

Documentation should use relative links and preserve the canonical FR, QR, CON, UC, CH, ADR and risk IDs.

For remaining P2 work, the proposed collaboration rule is one feature/section branch per coherent contribution, a pull request into main, a meaningful teammate review, and documented resolution before merging. This process is being introduced during P2; do not report past direct-to-main commits as PRs. Each person's row in Section 0.1 must link to actual contribution evidence. The docs/p1/README.md change addressing P1 C4 feedback was merged in PR #9.

The P2 submission requires one PDF containing Sections 0–12 and supporting diagrams, a commit containing that PDF and editable diagram sources, and the `p2` Git tag pointing to the PDF commit. Verification of the tag, instructor access, and Canvas upload remains pending. The deadline is November 8, 2026, 11:59 PM Central.

10.5 Use of AI tools in P2

The following table records known AI-tool usage by the listed contributors. Disclosures from other team members have not yet been confirmed.

Member	Tool	What it was used for	What you checked or changed
Nicholas Suther	Claude (Anthropic)	Building the spike S-01 code and drafting the README, ADR-02, the FR change rows, the Section 3 updates (including FR-16) and the UC-02 expansion; inserting Sections 3 and UC-02 into the shared document	Ran the spike myself and updated the results with my own measured times; checked results against FR-01, FR-02 and FR-08; edited wording
Patrick Paul	ChatGPT (OpenAI)	Assisted with drafting and revising P2 Sections 1, 2, and 10; checking FR/CH consistency; preparing the P3 plan, risks, PR evidence, and the P2 document update.	Checked repository PRs and synchronized P2 sections with the approved Table 3.1 and the still-open Q3/Q9 questions. Final exported-PDF layout review remains pending.
Patrick Paul	Qwen3.8-Flash-Next via Strata	Locally read repository and P1/P2 materials, including scope, project plan, CH changes, ADR-02, and spike S-01; helped identify missing P2 tasks and prepare draft wording.	Compared suggestions with the P1 baseline and CH-01–CH-03; retained the weight/thickness scope change and treated S-01 as simulated evidence, not actual-scale verification.

Member	Tool	What it was used for	What you checked or changed
Patrick Paul	Hermes Agent (Nous Research)	Reviewed and modified P2 documentation, performed Git operations, and compared Sections 1, 2, and 10 against the P1 baseline and reviewer feedback.	Patrick verifies documentation changes and Git operations before final export.
Ahmet Keles	Claude (Anthropic)	Drafting section 6, ADR-01, ADR-03, the UC-03 analysis (4.2, 4.3, 5.2.2) and section 11 from our P1 baseline and repository; generating the component, deployment and UC-03 sequence diagram sources; Git and PR guidance	Read each draft before committing it under my own account (PR #12); asked for corrections to the UC-03 wording; confirmed on GitHub that my commits show under ahmet-keles; read Peter's stakeholder interview before approving PR #13
Peter & Lucas	ChatGPT (OpenAI)	Formatting for section 5 UML Diagram and table	Checked against UML diagram from class slides, also cross referenced class and attributes to P1.
Lucas Pham	Claude	Double check for missed errors and used for formatting on md files in Git.	Checked reported errors used for section 5.3

11 Open Questions (updated; was P1 section 5)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q1	What weight and thickness limits decide the packaging type.	P1 requirements	Peter, Nov 18	Answered 11/05 by the owner by email (Stakeholder_Interview, PR #13): Table 3.1 thresholds approved; valuable-book request added as FR-16 (CH-05)	Table 3.1 as approved
Q2	Is the inspector a different person from the packer, and always available?	P1 (A3)	Ahmet, Nov 18	Open	A separate inspector during business hours

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q3	Does the existing scale connect to the PC, and what does it send?	P1 (A2, CON-03)	Nicholas, Nov 18	Partly answered: S-01 works on a simulated scale; real scale untested	Serial line format as in S-01, manual fallback (ADR-02)
Q4	What happens to a book that fails inspection?	P1 (UC-03)	Ahmet, Nov 18	Open	Returned for repackaging through UC-01 (UC-03 A2)
Q5 (new)	How does BookPack reach the existing shipping record system (API, database, export) and with what account?	Section 6.4 deploym ent	Ahmet, Nov 25	Open	HTTPS over the store LAN, behind Record Gateway
Q6 (new)	Does the inspector use a packing-station PC or a separate computer?	Section 6.4 deploym ent	Ahmet, Nov 18	Open	Station PC in the inspector role
Q7 (new)	Does the bookstore accept that a book read at exactly 5.0 lb on the scale may be flagged (CH-01)?	Spike S-01 / R7	Peter, Nov 18	Open	Extra inspection is acceptable
Q8 (new)	May a packing employee override the reinforced box that a valuable book receives (FR-16)?	UC-02 analysis	Peter, Nov 18	Open	The override is allowed but requires approval from the bookstore owner or manager.
Q9 (new)	Is 5.0 lb the right flag limit?	P1 requirem ents	Peter, Nov 18	Open	5.0 lb flag limit as written; thresholds in rules.json (ADR-03)

12 Baseline v1.0 Declaration (was P1 section 6)

We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date
Peter Yu	Peter Yu	11/08/2026
Ahmet Keles	Ahmet Keles	11/08/2026

Team member	Signature or initials	Date
Nicholas Suther	Nicholas Suther	11/08/2026
Patrick Paul	Patrick Paul	1108/2026
Afeef Zarraf	Afeef Zarraf	11/08/2026
Lucas Pham	Lucas Pham	11/08/2026
Jasmine Tran	Jasmine Tran	11/08/2026